



SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED
AUTONOMOUS AIR SYSTEMS • SWARM INTELLIGENCE • TACTICAL LOGISTICS

MADE IN INDIA • ATMANIRBHAR BHARAT

Engineering Collaboration: BCN In-House Technology Team

PROGRAM BLUEPRINT & TABLE OF CONTENTS

CAPABILITY ARCHITECTURE & TECHNICAL INDEX

A comprehensive operational reference detailing sovereign autonomous swarm intelligence, heavy-cargo hybrid VTOL logistics, and frontline border defence architectures engineered to Indian Armed Forces requirements.

12 TECHNICAL SECTIONS

3 CORE CAPABILITY PILLARS

210+297 MM A4 MASTER BROCHURE

★ MAKE IN INDIA & IDEX COMPLIANT

PART I • TACTICAL UAS

SWARM & OVERWATCH ISTAR

AUTONOMOUS SWARM INTELLIGENCE & CONTESTED MESH C2

01	EXECUTIVE VISION & DOCTRINE	PAGE 03
Forward Border Operations • Multi-Tier Defence High-altitude border operational requirements and multi-tier sovereign air defence integration doctrine.		
02	HETEROGENEOUS SWARM C2	PAGE 04
Peer-to-Peer Mesh • Autonomous Patrol Distributed peer-to-peer RF mesh routing, decentralized patrol logic, dynamic collision avoidance, and swarm target handoff.		
03	NANO-SCOUTS & OVERWATCH RELAYS	PAGE 05
Micro Scouts (350g) • 20kg Class Relays Disposable micro air vehicles teamed with long-endurance Overwatch relay nodes for continuous multi-spectral EO/IR surveillance.		
04	SORA SAFETY & OBCA COMPLIANCE	PAGE 06
SAIL II/III Assessment • FTS Mitigation Specific Operations Risk Assessment (SORA SAIL II/III), emergency termination systems, and sovereign military airspace clearances.		

PART II • CARGO VTOL

TACTICAL LOGISTICS VTOL

RUNWAY-FREE FORWARD RESUPPLY & COLD-CHAIN TRANSIT

05	HYBRID AIRFRAME ARCHITECTURE	PAGE 07
VTOL • Fixed Wing • High-Altitude Lift Vertical lift rotor transition to high-efficiency fixed-wing cruise engineered for forward mountain posts without prepared runways.		
06	AMMUNITION & SUPPLY TRANSIT	PAGE 08
5-10kg to 200kg • 100+ km Mission Radius Forward line resupply handling vital stores, tactical munitions, and spare parts across 100+ km radius in zero-infrastructure terrain.		
07	TRAUMA CARE & PRECISION WINCH	PAGE 09
Blood Cold-Chain • Winch & Air-Drop Temperature-controlled medical pods for whole blood and antidote delivery, featuring autonomous winch delivery and parachute failsafe.		
08	HEAVY-LIFT HYBRID POWERTRAINS	PAGE 10
Hybrid Fuel/Electric • 180 Min Endurance Multi-fuel electric hybrid propulsion, sub-zero battery thermal conditioning, and 180-min endurance in Himalayan mountain wind gradients.		

PART III • MAKE IN INDIA

SOVEREIGN INFRASTRUCTURE

COMMAND & CONTROL, C-UAS HARDENING & PRODUCTION

09	UNIFIED TACTICAL GCS CONSOLE	PAGE 11
Common C2 • Bust Swarm & Cargo Control Ruggedized military workstation controlling simultaneous swarm reconnaissance and cargo VTOL sorties from a single tactical screen.		
10	EW & NAVIC TELEMETRY HARDENING	PAGE 12
Anti-Jam Frequency Hopping • NAVIC Sync Anti-jamming frequency-hopping RF data-links, sovereign NAVIC satellite synchronization, and optical dead-reckoning fallback under EW denial.		
11	INDIGENOUS MANUFACTURING PLAN	PAGE 13
Composite Autoclave • Localized Supply Sovereign carbon-composite fabrication, localized component assembly, and IDEX / MoD procurement roadmap for scalable Indian production.		
12	CORPORATE & DEFENCE CONTACT	PAGE 14
CIN Registration • Technical Collaboration Company legal credentials, soft in-house technology collaboration structure, field demonstration schedule, and user engagement contact.		

SOVEREIGN DOCTRINE

UNIFIED AIR ARCHITECTURE

One tactical ecosystem linking micro swarm scouts, overwatch relays, and heavy cargo VTOLs under synchronized command.
• STANAG 4586 Interoperable

ALL-TERRAIN VTOL

RUNWAY-FREE ALTITUDE ENVELOPE

Vertical departure and transition to 180 km/h fixed-wing cruise, certified for harsh sub-zero temperatures up to 16,000 ft AMSL.
• Sub-Zero Himalayan Tested

CONTESTED SPECTRUM

NAVIC • MESH ENCRYPTION

Resilient anti-jam frequency hopping with autonomous optical dead-reckoning fallback under intense GNSS spoofing or jamming.
• Dual-Band NAVIC • INS

MAKE IN INDIA

100% SOVEREIGN IPR & MRO

Domestic carbon-composite production and localized depot-level lifecycle maintenance aligned with Defence Acquisition Procedure (DAP).
• IDEX / AITI Compliant

MULTI-DOMAIN UNMANNED AERIAL ECOSYSTEM • DEFENCE SCALING ROADMAP

Phased Deployment: Prototyping & High-Altitude Trials → MoD User Evaluation → Sovereign Induction

12 TECHNICAL SECTIONS

3 OPERATIONAL PILLARS

★ TRL 7+ TESTED

■ SOVEREIGN IPR